

Research article

Diagnosis of epileptic seizures from electroencephalogram signals using log-Mel spectrogram and a deep learning CNN model

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ABSTRACT

The traditional procedure of the electroencephalogram (EEG) examination is laborious, slow, and prone to many diagnostic mistakes, creating a need for automated systems capable of reliably detecting epileptic seizures or other significant neurological conditions. To address this issue, this study proposes a novel deep learning-based method for automating seizure detection using the log-Mel spectrogram of EEG signals. The method innovatively transforms EEG data into images by applying the logarithm to the Mel-filter bank spectrograms of the signals, allowing convolutional neural networks (CNNs) to classify the resulting image representations. A CNN model is designed and trained on these images for classifying EEG signals. The proposed method is evaluated on two publicly available datasets, for both binary and multi-class classifications. Experimental findings on the Bonn dataset show that the proposed method's accuracy, sensitivity, specificity, precision, and F₁-score for five-class classification are 98.13 %, 95.33 %, 98.83 %, 95.59 %, and 95.32 %, respectively. For the three-class classification on the same dataset, the accuracy, sensitivity, specificity, precision, and F₁-score are 99.60 %, 99.33 %, 99.67 %, 99.52 %, and 99.41 %, respectively. For the three-class classification on the NSC-ND dataset, accuracy, sensitivity, specificity, precision, and F₁-score are 93.33 %, 90.00 %, 95.00 %, 91.89 %, and 89.50 %, respectively. The proposed approach outperforms existing seizure detection methods, demonstrating superior classification performance. To the best of our knowledge, it is the first to use log-Mel spectrograms of EEG signals for epileptic seizure classification. These findings highlight the effectiveness of combining log-Mel spectrograms with CNNs for EEG-based epilepsy seizure detection, showcasing its potential for real-world automated epilepsy diagnosis.

1. Introduction

Over 50 million people worldwide suffer from epilepsy, a neurological disorder [1]. Epilepsy is characterized by a brain's tendency to produce unprovoked epileptic seizures, which are spontaneous seizures that cannot be immediately attributed to any specific cause [2]. EEG data must be interpreted in order to diagnose epilepsy. Visual inspection of the EEG waveform is the most common method used in medical interpretation of EEG information. It takes a lot of time to complete this practice. The EEG signal has numerous

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applications in biomedical fields such as brain-computer interface [3], disease detection [4–5], emotion recognition [6], operator functional [7], state sleep stage classification [8], person identification [9], age and gender prediction [10]. The proposed method focuses on disease detection, specifically identifying epileptic seizures. EEG waveforms are central to the gold standard diagnostic procedure for confirming epilepsy. Studies have reported that the sensitivity of this procedure for identifying inter-ictal epileptiform discharges in epilepsy centers ranges from 29 to 55 % [11]. Identifying the type of epileptic seizure is critical for understanding the brain's cortical connectivity and for neurosurgical planning [12]. This identification is also crucial for managing epilepsy patients' conditions and treatment [13]. Thus, there is an urgent need for an automated, reliable, and efficient real-time system for detecting epileptic seizures.

Numerous techniques have been proposed to automatically identify epileptic seizures in EEG signals, eliminating the need for lengthy visual inspection [14]. Typically, these techniques involve two steps: feature extraction and classification. The feature extraction methods used in previous studies include: log energy and norm entropy [15], matrix determinant [16], filter bank [17], wavelet packet log energy and norm entropies [18], joint time-frequency analysis [19], non-linear analysis, wavelet analysis [20–23], and time and frequency domains analysis [19]. Both supervised and unsupervised approaches have been proposed for the classification step. In Ref. [24], the authors proposed using a support vector machine (SVM) as a classifier, with a successive decomposition index as features, to identify epileptic episodes in long-term EEG recordings. They also assessed the effectiveness of the SVM classifier for automated seizure detection using features derived from the discrete wavelet transform-based sigmoid entropy [25]. Mumtaz et al. [26] proposed using wavelet decomposition in combination with hidden Markov models for classifying EEG waveforms. In Ref. [27], the authors employed kernel-based methods to distinguish between focal and non-focal EEG waveforms using entropy characteristics based on Jacobi polynomial transforms. In Ref. [28], the authors employed wavelets for feature extraction, and used associative Petri nets and fuzzy entropy to detect epilepsy. Wavelets, principal component analysis (PCA), artificial neural network (ANN), and SVM were employed by Atemangho et al. [29] for the detection of epileptic seizures. To identify epileptic seizures and ocular states, Djoufack et al. [30] employed polynomial-based rhythm decomposition for extraction, linear discriminant analysis (LDA) for discriminative features, and SVM for classification. The same authors used PCA, kernel PCA (KPCA), LDA, and gaussian discriminant analysis (GDA) with kernel machines to compare feature extraction techniques based on polynomials for the same goal [31]. Using wavelet packet decomposition and the quad binary pattern method, Sairamya et al. suggested a method for automatically detecting focal EEG signals [32]. By employing key-point-based local binary patterns of EEG waveforms and their histogram as features to feed the SVM for EEG waveform recognition, Tiwari et al. [33] were able to identify epilepsy. For the purpose of detecting epileptic seizures, Gupta and Pachori [34] employed weighted multi-scale Renyi permutation entropy (WMRPE) and rhythms extracted using Fourier-Bessel series expansion (FBSE) of EEG signals to feed different classifiers like random forest (RF), regression, and least squares support vector machine (LS-SVM). To increase the accuracy of epilepsy diagnosis, Subasi et al. [21] applied a hybrid approach combining an SVM classifier, particle swarm optimization (PSO), and a genetic algorithm. Shankar et al. [35] proposed using recurrence plot images as features for deep learning models to diagnose epileptic seizures from EEG signals [35]. Bhattacharyya et al. [36] suggested fusing the signal decomposition with k-Nearest Neighbors (k-NN) entropy and tunable-Q wavelet transform (TQWT) filtering as features for the SVM classifier. Zahra et al. [37] used the multivariate empirical mode decomposition (MEMD) technique to distinguish between ictal and non-ictal EEG waveforms. While these techniques have shown promise in seizure identification, they still face limitations in terms of scalability and the ability to generalize across diverse datasets. Furthermore, many conventional approaches may not adequately capture both high- and low-frequency components of EEG signals, which are crucial for accurate seizure detection.

Because of the accuracy of the classification results they produce in the fields in which they are utilized, deep learning architectures are currently by far the most popular classifiers. The authors of [14] give an overview of current advancements and offer predictions for the future of deep learning in EEG. A 13-layer deep convolutional neural network (CNN) technique was proposed by Acharya et al. [38] for the detection of normal, pre-ictal, and seizure classes. To identify epileptic seizures, Omar and El-Hafeez employed nine deep learning models with various preprocessing methods [39]. The authors of [40] suggested utilizing stacked long short-term memory (LSTM) and bidirectional LSTM (Bi-LSTM) to identify seizures. A deep learning technique based on stacking ensembles was proposed by Akyol [41] to determine whether or not epileptic seizures are occurring. The CNN was taken into consideration in the study of [42] to improve the effectiveness of the epileptic seizure recognition method. The CNN and the transfer learning method were proposed by Temel et al. [43] for the detection of epileptic seizures. Zubair et al. integrated machine learning models with dimensionality reduction strategies [44]. Using intracranial EEG recordings, Geng et al. [45] presented the Stockwell transform (S-transform) and Bi-LSTM neural networks for epileptic seizure detection. A deep learning technique for automatic seizure identification from long-term scalp EEG utilizing the spectrogram was proposed by the authors in Ref. [46]. Chowdhury et al. [47] used a 1-D CNN architecture to identify seizures and non-seizures in EEG signals. A mixed deep learning model was suggested by Varli et al. [48] to automatically identify epileptic seizure activity. By merging multiple models, including the Gated Recurrent Unit (GRU) and Bi-LSTM, Mallick et al. [49] were able to identify epileptic seizures. Using a pre-trained model as a classifier, Kaur et al. [50] employed a visual representation of an EEG signal created by applying the wavelet transform (WT) at various scales and time intervals as features. The retrieved features are then run through SVM to classify the EEG waveform. To distinguish preictal EEG from normal EEG, Yap et al. [51] created a number of supervised and unsupervised deep learning models. Turk et al. [52] employed a CNN structure to learn the features of the EEG waveform's scalogram images.

While deep learning architectures, particularly Convolutional Neural Networks (CNNs), have demonstrated great success in image classification tasks, their application to EEG signal classification has been relatively limited. Some studies have employed CNNs in combination with spectrograms or scalograms, which represent EEG signals in the time-frequency domain. However, existing methods often rely on conventional spectrogram representations, which may not fully exploit the temporal and spectral intricacies of EEG

signals. This paper addresses these gaps by proposing a novel deep learning-based method for epileptic seizure detection, utilizing log-Mel spectrograms (LMS) as input features. LMS, a time-frequency representation commonly used in audio processing [53–58], is particularly suited for capturing both temporal and spectral characteristics of EEG signals. By leveraging LMS for feature extraction, we aim to enhance the representation of EEG data, making it more suitable for deep learning models, particularly CNNs, which excel at learning hierarchical feature representations. The proposed method introduces an innovative approach for seizure detection, where the log-Mel spectrograms are used as input to a CNN model, providing an end-to-end solution for automatic seizure classification. This approach simplifies the complex task of detecting epileptic seizures from EEG data, offering an efficient and scalable alternative to traditional methods. This research is of significant academic value as it advances the application of deep learning techniques to the seizure detection problem and addresses the limitations of previous methods. By utilizing log-Mel spectrograms, this work introduces a novel feature extraction technique that enhances the model's ability to accurately capture the complex temporal and spectral patterns inherent in EEG signals.

The remainder of this paper is organized as follows: Section 2 reviews related work on epileptic seizure detection using EEG signals; Section 3 outlines the materials and methods used in this study; Section 4 presents the results and discusses the findings; and Section 5 concludes the paper and outlines potential directions for future research.

2. Related works

Neural network-based algorithms have been suggested in several publications as a solution to the diagnosis of epileptic seizures [14,38–52]. CNN models are particularly advantageous since they can automatically extract features. Numerous authors classified EEG data using deep learning techniques based on 2D CNNs, especially when diagnosing epileptic seizures. EEG image representations are used as features in these techniques. Time-frequency or time-scale features are frequently extracted during the conversion of the 1D EEG data into a 2D image. Several databases have been presented for the performance evaluation of epileptic seizure detection systems on EEG signals. For this purpose, Wong et al. [59] conducted a review of the databases used in the prediction and detection of epileptic seizures.

The multivariate empirical mode decomposition (MEMD) approach was employed by Zahra et al. [37] to identify seizures from EEG signals. An input EEG signal is broken down into distinct frequency bands using the empirical mode decomposition technique. The multiscale time frequency representation is used by the authors to choose appropriate feature sets. These characteristics were fed into an ANN classifier that detects seizures from EEG recordings. This method's accuracy for a five-class problem on the Bonn dataset was 87.2 %, according to the evaluation. The features at various frequencies and time intervals were extracted in Ref. [50] using the wavelet transform. The authors then employ the pre-trained AlexNet network model with this kind of feature, known as a scalogram. New features are taken from certain network layers after the network has been trained so that the SVM classifier (support vector machines) can use them. The diagnostic result is produced using the SVM classifier. Given the number of classifiers that make up this model, it appears to be rather sophisticated. It was evaluated on signals from two EEG databases: the Bonn University database [60] and the NSC-ND for the Hauz Khas Center for Neurology and Sleep in New Delhi [61]. By altering the wavelet type, the scientists were able to attain binary classification accuracy ranging from 98.5 % to 100 % on the Bonn database. An accuracy of 98 %–98.8 % was achieved when the five categories were divided into three groups. After dividing the three classes into two groups, the authors were able to attain an average accuracy of 98 %–100 % on the NSC-ND database. To improve classification performance in the identification of epileptiform patterns, Swami et al. [62] suggested choosing frequency subsets. To acquire features, they employed the dual-tree complex wavelet transform (DT-CWT) with six decomposition layers. For the classification, the authors fed a general regression neural network (GRNN) with these features. These authors evaluated their approach by separating epileptic seizures from non-seizures. They achieved 98.1 % accuracy, 99.4 % sensitivity, and 99.7 % specificity on the NSC-ND dataset. To detect epileptiform patterns in EEG recordings, the authors of [63] compared wavelet packet transform, discrete wavelet transform (DWT), and dual-tree complex wavelet transform (DTCWT). The general regression neural network classifier receives the extracted features as inputs in the form of energy (ENE), root-mean-square values, and standard deviations (STD). They were able to classify non-seizures from epileptic seizures using the NSC-ND dataset. The same group of authors developed an automated classification system requiring minimal supervised training [64]. They decomposed signals into time-frequency subbands up to the sixth level using DTCWT. Features such as energy, standard deviation, root-mean-square, Shannon entropy, mean values, and maximum peaks were extracted from the detail and approximation coefficients. These features were classified using a GRNN with K-fold cross-validation under different train-to-test ratios. Tzallas et al. [5] introduced an alternative to the classical spectrogram for EEG time-frequency analysis. They used short-time Fourier transform and various time-frequency (t-f) distributions to calculate the power spectrum density (PSD) of each segment. The process involves three stages: t-f analysis and PSD calculation, feature extraction based on fractional energy in specific t-f windows, and classification of the EEG segment (seizure detection) using artificial neural networks. Acharya et al. [4] used discrete wavelet transform to decompose EEG signals into frequency sub-bands. The resulting wavelet coefficients were processed with independent component analysis (ICA) for dimensionality reduction. These features were then classified using six classifiers: decision tree, K-nearest neighbor, probabilistic neural network, fuzzy, Gaussian mixture model, and support vector machine, to identify the best-performing classifier. In order to classify EEG signals for the diagnosis of epilepsy, the authors of [19] suggested using the spectrogram image as features to train a 2D CNN model. The Short Time Fourier Transform (STFT) of EEG signals is calculated to produce the spectrogram. Three convolution layers, ReLU and max pooling layers, three fully connected layers, a softmax layer, and a classification layer make up their CNN model. Only EEG signals from the Bonn database were used by these scientists to test their model. Their binary classification accuracy ranged from 98.88 % to 100 %. The accuracy ranges they achieved with the three-class categorization were 97.33 %–99.33 %. Five distinct adaptive decomposition techniques were evaluated by Carvalho et al. for the

purpose of detecting and classifying seizures in EEG signals [65]. They trained classifiers like the radial basis function (RBF), the support vector machine (SVM), the k-nearest neighbors (KNN) with three neighbors, the Gaussian process classification (GPC), and the multi-layer perceptron (MLP) using features from empirical mode decomposition (EMD), extended EMD (EEMD), complete EEMD with adaptive noise (CEEMDAN), variational mode decomposition (VMD), and empirical wavelet transform (EWT). The authors tested their algorithms on the three-class classification (A-D-E) on the Bonn database. They were particularly interested in the Normal-Ictal and Preictal-Interictal-ictal classification difficulties on the NSC-ND database. The use of spectrograms and scalograms to translate EEG signals into images was recently proposed by Varli et al. [48]. The RNN (Recurrent Neural Network) and CNN, two models of the integrated deep learning architecture, were then used to assess these image attributes independently. They achieved minimum and highest accuracy rates of (99.07 %, 99.28 %) for binary classifications and (97.60 %, 98.56 %) for triple classifications on the Bonn dataset.

Based on current research, there is still room for improvement in the seizure detection systems' performance outcomes. The majority of the techniques have been tested on a single database, primarily the Bonn database, which appears to be the most popular for using EEG signals to diagnose epileptic seizures. Additionally, the classification in the five fundamental categories A, B, C, D, and E of the Bonn dataset does not really interest many authors. After grouping, they are frequently restricted to two or three classes. One of the major challenges in seizure detection is how to effectively represent EEG data. Traditional methods rely on manually designed features that may not capture the full complexity of EEG signals. Deep learning models like CNNs can learn features automatically, but they often struggle with the raw time-domain signals of EEG, requiring sophisticated preprocessing techniques. For feature extraction, time-frequency and time-scale transformations are frequently employed. The benefit of this kind of feature is that CNN networks can classify images quite well. One kind of feature derived from the spectrogram in the Mel scale is the Mel spectrogram. This is accomplished by using Mel-filter banks to calculate the spectrogram. A Mel bin is created by grouping many short-term Fourier transform frequencies into a Mel spectrogram [66]. As with audio signals, we believe that combining multiple STFT frequencies into a Mel bin can alter the features and probably increase their efficacy in classifying EEG signals [53–58]. The proposed method employs a CNN as a deep learning model for the classifier. This is the solution that provides improved performance together with ease of implementation.

3. Materials and methods

Fig. 1 shows the synoptic diagram of the proposed method. It follows many steps such as converting the EEG signal into images via the logarithm of the time-frequency spectrogram on the Mel scale. The obtained images are used to feed a CNN model to detect or recognize the presence of an epileptic seizure in the considered EEG signal.

3.1. EEG data description

EEG signals used in this work were extracted from two EEG datasets freely available online: one is from the University of Bonn [60], and another is from the Neurology and Sleep Center, Hauz Khas, New Delhi (NSC-ND) [61]. They are publicly available databases generally used in the classification of EEG signals for the diagnosis of epileptic seizures.

A University of Bonn dataset

It is the database of Department of Epileptology from university of Bonn, Germany [60]. This dataset is commonly used as a

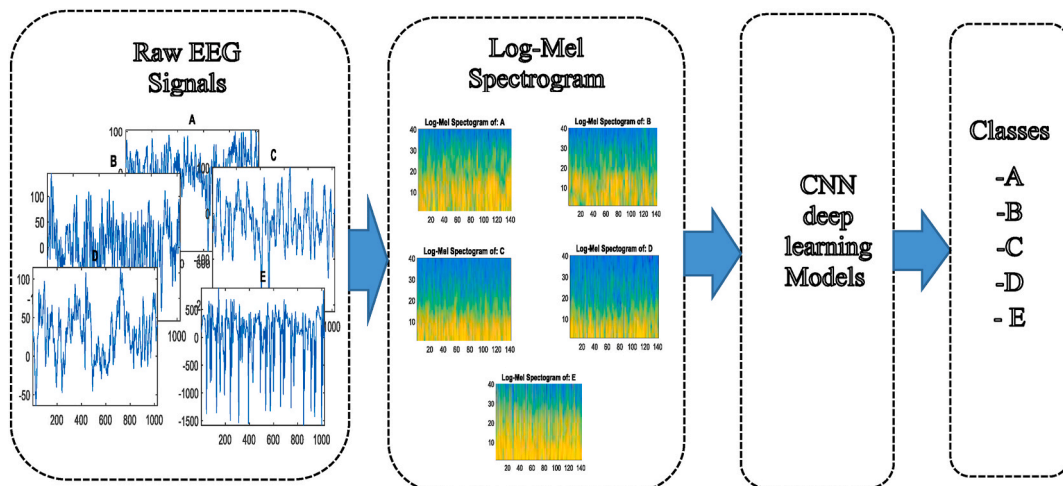


Fig. 1. Block diagram of the proposed method.

benchmark for seizure detection algorithms. These EEG signals were recorded using the 10–20 international electrodes positioning system [52]. They were filtered through a 0.53–40 Hz bandpass filter, with a 173.6 Hz of sampling rate. EEG samples have a duration of 23.6 s with 4097 recording points. This dataset contains five sets of EEG signals (denoted A–E). Each set contains 100 single-channel EEG segments. Each segment was obtained by selecting and cutting from continuous multichannel EEG recordings, following visual inspection for artifacts [60]. The sets A and B are recorded by surface EEG modality from five healthy volunteers. The set A is that of persons with eyes open. The set B is that of persons with eyes closed. The sets C, D and E are acquired by intracranial modality on epileptic patients. The set C is recorded during the hippocampal formation of the opposite hemisphere of the brain of the patient in a seizure-free interval. The set D is the recording of epileptogenic zone in a seizure-free interval. The set E is acquired during seizure activity. Fig. 2 shows an EEG curve for each of the concerned sets. Each set corresponds to a class of EEG waveform.

B Dataset of the Neurology and sleep center - New Delhi (NSC-ND)

This dataset contains segmented EEG signals recorded from epilepsy patients at the Neurology and Sleep Center, Hauz Khas, New Delhi. Each EEG recording is digitized with a sample frequency of 200 Hz, filtered from 0.5 Hz to 70 Hz. Each segment has a duration of 5.12 s [64]. Each segment is associated with one of the following three categories corresponding to the epileptic seizure stage: preictal, interictal, and ictal. Fig. 3 presents a curve for each of the concerned classes. This dataset contains 150 segments (50 per class) [67].

3.2. Log-Mel spectrogram based feature extraction

The log-Mel spectrogram (LMS) is the log-transformed version of the Mel scaled spectrogram. It is a method to extract spectral information from a signal. We wish to use this technique to extract features from the EEG signal in order to differentiate the state of an epileptic seizure compared from other states. An EEG signal consists of the values of a non-stationary time function whose frequency varies over time. The properties of changes on this type of signal cannot be fully described using only the information in the frequency domain. The spectrogram is a 2D time-frequency information obtained from the short-time Fourier Transform (STFT) of a 1D time signal.

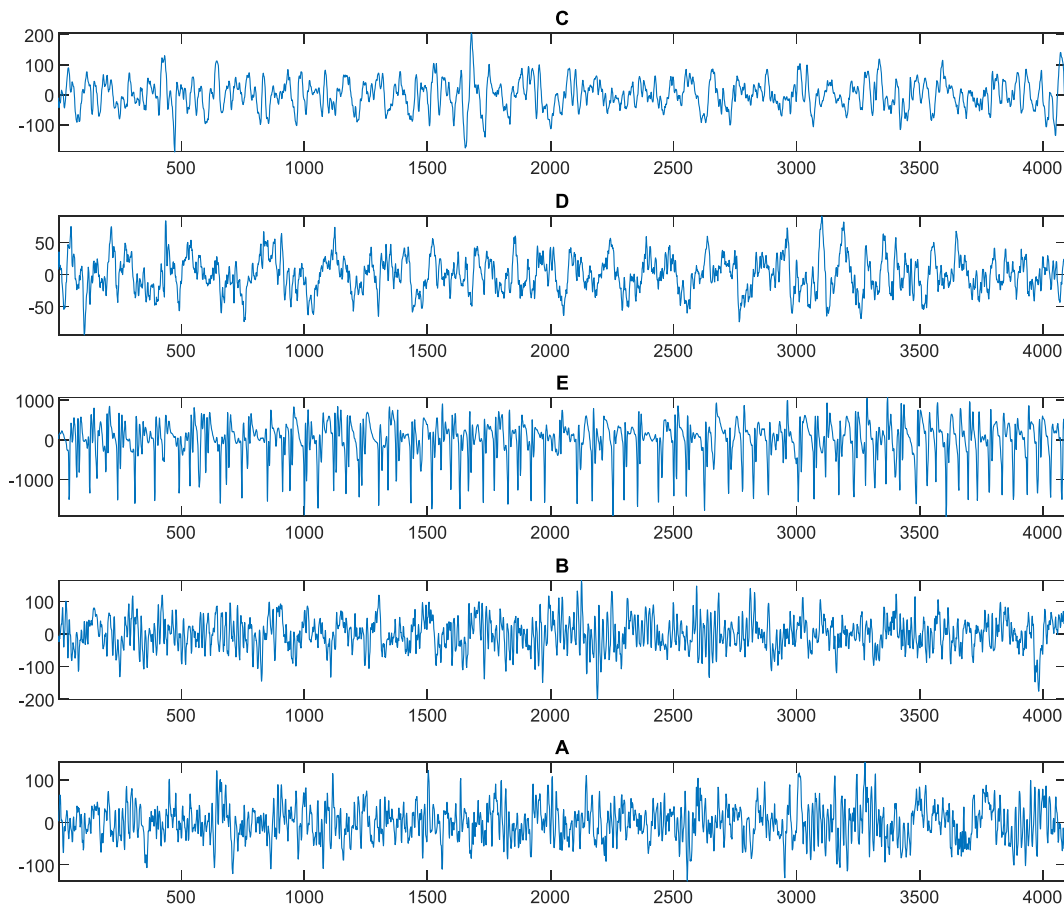


Fig. 2. Waveforms of different sets of EEG signals used from the University of Bonn dataset. From top to bottom, we have, respectively, the classes C, D, E, B and A.

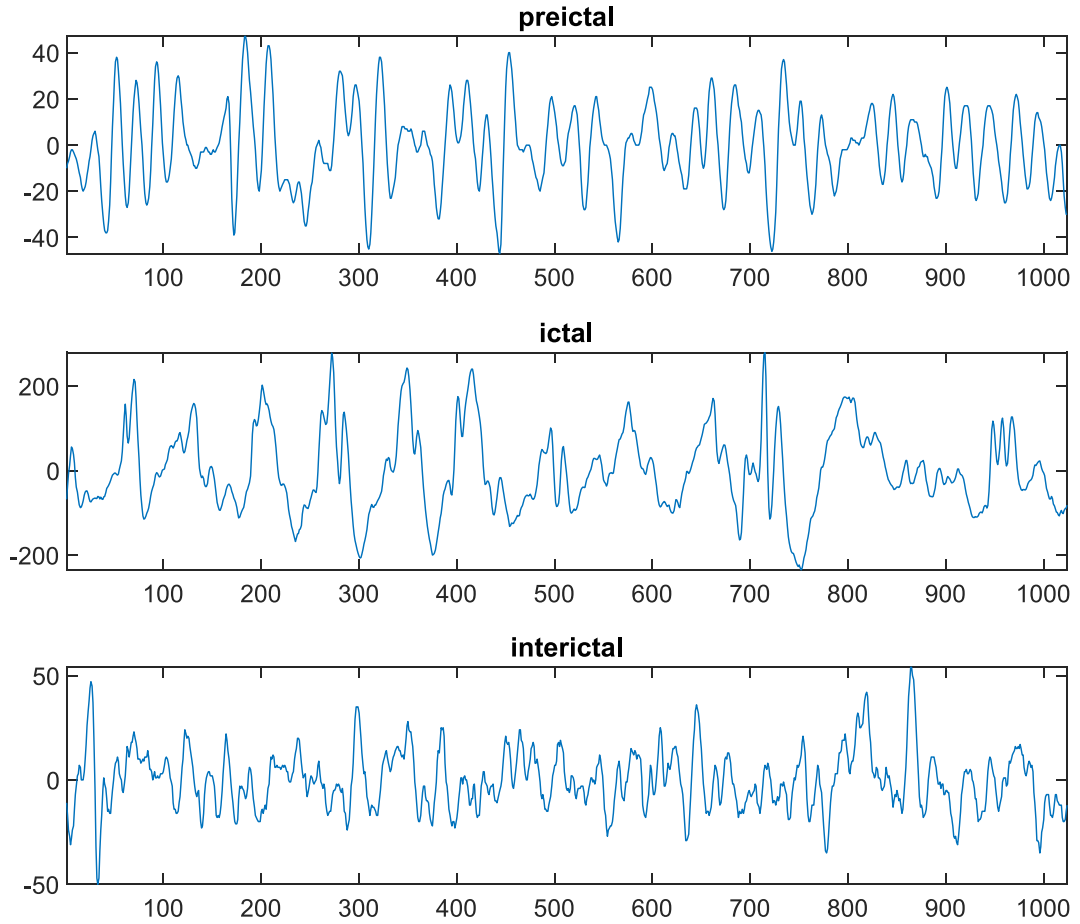


Fig. 3. Waveforms of different sets of EEG signals used from the NSC-ND database.

Let us consider a discretized digital signal $x[n]$. $x[n]$ represents an EEG signal. Its time-frequency spectrogram is defined by the following equation (1), for the k th frequency bin and the a th time-frame [56].

$$STFT\{x[n]\} = X(a, k) = \sum_{n=0}^{N-1} x[n] \cdot w[n - aH] \cdot e^{-j2\pi kn/N} \quad (1)$$

$0 \leq k \leq N-1$. $w[n]$ represents a window function with length L . H is the hop size. N is the total number of frequency bins.

The computed spectrogram passes through Mel-filter banks to convert the spectrogram into Mel scale. The obtained result is known as the Mel spectrogram. It groups multiple STFT frequency bins into one Mel bin. The log-Mel spectrogram (LMS) coefficient for the a th frame and the m th Mel-filter bank, is given as follows (2):

$$LMS\{x[n]\} = LMS(a, m) = \log \left[\sum_{k=0}^{N-1} H_m[k] \cdot |X(a, k)|^2 \right] \quad (2)$$

$0 \leq m \leq M-1$, where $H_m[k]$ is the k th coefficient for the m th filter bank defined as follows (3) [55–57].

$$H_m[k] = \begin{cases} \frac{k - f[m-1]}{f[m] - f[m-1]}, & f[m-1] \leq k \leq f[m] \\ \frac{f[m+1] - k}{f[m+1] - f[m]}, & f[m] \leq k \leq f[m+1] \\ 0, & \text{otherwise} \end{cases} \quad (3)$$

$f[m]$ is the Mel bin point given by (4) [61]

$$f[m] = \left[\frac{N+1}{F_s} B^{-1} \left(f_{mel}^l + m \frac{f_{mel}^h - f_{mel}^l}{M+1} \right) \right] \quad (4)$$

F_s is the sampling rate, M is the number of Mel-filter banks, f_{mel}^h and f_{mel}^l are highest and lowest frequencies of the filter banks in Mel, and B^{-1} is an operator that converts Mel into Hertz [56].

$$B^{-1}(f) = (700 \cdot (10^{f_{mel}/2595} - 1)) \quad (5)$$

Fig. 4 presents examples of the obtained images for each class from both databases used. The first row shows the raw 1D EEG signal. The second row is the corresponding spectrogram. The third and fourth row show, respectively, the corresponding images of the Mel-spectrogram and the log-Mel spectrogram. The Log-Mel Spectrograms feature extraction method includes the following steps.

- Windowing: the EEG signal is divided into overlapping segments or windows. The window (Hanning window) length L and hop size H are chosen to balance time resolution and frequency resolution. These parameters are defined as $L = 102$ samples and $H = 41$ samples in this experiment for the Bonn Dataset, with $N = 4097$. For the NSC-ND dataset, with $N = 1024$, these parameters are $L = 28$ samples and $H = 10$ samples. Each window represents a portion of the EEG signal from which a spectrogram will be derived.
- Mel-Filter Bank Application: A Mel-filter bank is applied to each windowed segment. By using Mel-filtering, the EEG signal is transformed into a time-frequency representation that captures its key features in the Mel frequency domain. For the Bonn Dataset, with $F_s = 173.6$ Hz, the number of Mel-filter banks is $M = 40$, frequency range = [0 Hz, 86 HZ]. For the NSC-ND dataset, with $F_s = 200$ Hz, $M = 40$ and frequency range = [0 Hz, 100 HZ].
- Logarithmic Transformation: After applying the Mel-filter bank, the logarithm of the Mel spectrogram is computed to compress the dynamic range of the signal and enhance features that may be more relevant for classification. This logarithmic transformation produces a Log-Mel Spectrogram (LMS), which is then used as the input feature for the CNN.

The final output of the feature extraction step is a matrix (or image) where the rows correspond to the Mel frequency bins, and the columns correspond to time frames. This matrix, known as the LMS matrix, serves as the input for the classification step. For the Bonn dataset, the resulting image has dimensions of 40x98, while for the NSC-ND dataset, the image size is 40x100.

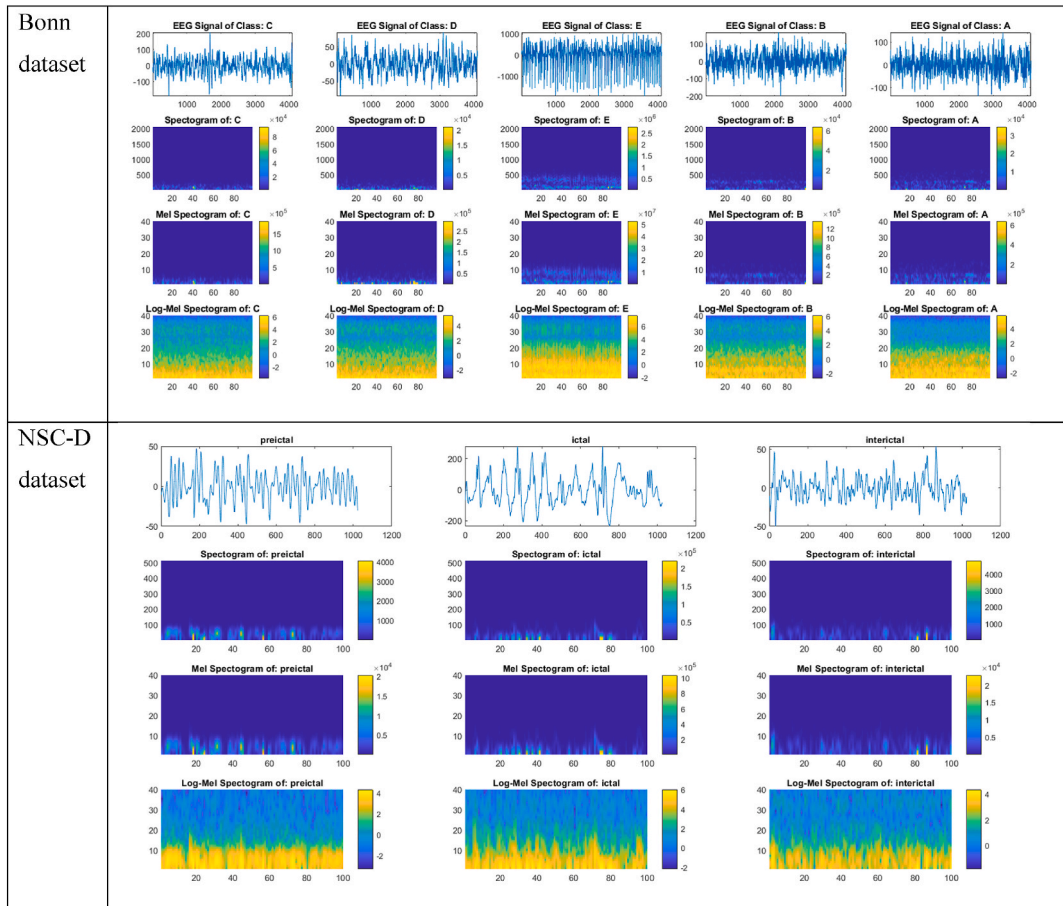


Fig. 4. Examples of the obtained images for each of five classes for the Bonn dataset and the three classes of the NSC-ND dataset. The first row shows the raw 1D EEG signal. The second, the third and fourth rows show, respectively, the corresponding images of the spectrogram, the Mel-spectrogram and the log-Mel spectrogram.

3.3. Convolutional neural network

As discussed in the previous paragraph, the log-Mel spectrogram produces an image, which is used as features for classifying different forms of EEG signals. Among image classifiers, CNNs is the most commonly used. There are many CNN models. Some of these models are specifically adapted to certain types of images with which they provide excellent performances. However, others can be quite complex and may not be suitable for simple applications intended for direct implementation. The goal is to use a simple and efficient model that can be easily implemented in an embedded system. For these reasons, we propose a new CNN model designed to adapt to this specific problem. The input to the CNN is the Log-Mel Spectrogram of the EEG signal. This spectrogram is essentially an image-like representation of the signal, where the x-axis represents time, and the y-axis represents frequency (in Mel scale). The spectrogram matrix contains the energy information of the EEG signal across both time and frequency domains, which is then fed into the CNN for classification. The input format to the CNN model is a matrix derived from the Log-Mel Spectrogram. This matrix representation of the signal is treated as a 2D image by the CNN. Each entry in the matrix represents the log-energy of a particular frequency at a particular time frame. *Convolutional neural network (CNN)* is a deep neural network architecture that learns feature by itself via filters. The CNN consists of various layers, including the convolution layer, pooling layer, fully connected layer, and dropout operations. The way these layers are connected allows for the creation of different CNN models. In this paper, we propose the use of a simple CNN model.

The architecture of the proposed network CNN model is made up of 34 layers of neurons with 33 connections. Table 1 shows the summary of the proposed CNN architecture along with the parameters for each layer. In this table, the layer parameter “zero-center normalization” refers to the process of subtracting the mean pixel value from each pixel of an image, so that the image’s pixel values are centered around zero. The following subsections describe the principle and role of each of these layers.

A. Image Input layer

It is the layer which receives the input images obtained from the phase of transforming the EEG signals into images.

B. Convolution layer

This layer performs the filtering operation in order to extract the most relevant features. The mathematical equation of the

Table 1
Summary of the proposed CNN architecture.

Layers	Type	Output Size	Layer parameters
Imageinput	Image Input	40x98x1	Images with “zero-center” normalization
batchnorm_1	Bach normalization	40x98x1	
Conv_1	2D Convolution	40x98x32	32 3x3 convolutions with stride 1
batchnorm_2	Bach normalization	40x98x32	
Relu_1	ReLU	40x98x32	
Conv_2	2D Convolution	40x98x32	32 3x3 convolutions with stride 1
batchnorm_3	Bach normalization	40x98x32	
Relu_2	ReLU	40x98x32	
Maxpool_1	2D max pooling	20x49x32	3x3 max pooling with stride 2
Conv_3	2D Convolution	20x49x64	32 3x3 convolutions with stride 1
batchnorm_4	Bach normalization	20x49x64	
Relu_3	ReLU	20x49x64	
Conv_4	2D Convolution	20x49x64	32 3x3 convolutions with stride 1
batchnorm_5	Bach normalization	20x49x64	
Relu_4	ReLU	20x49x64	
Maxpool_2	2D max pooling	10x25x64	3x3 max pooling with stride 2
Conv_5	2D Convolution	10x25x128	128 3x3 convolutions with stride 1
batchnorm_6	Bach normalization	10x25x128	
Relu_5	ReLU	10x25x128	
Conv_6	2D Convolution	10x25x128	128 3x3 convolutions with stride 1
batchnorm_7	Bach normalization	10x25x128	
Relu_6	ReLU	10x25x128	
Maxpool_3	2D max pooling	5x13x128	3x3 max pooling with stride 2
Conv_7	2D Convolution	5x13x256	256 3x3 convolutions with stride 1
batchnorm_8	Bach normalization	5x13x256	
Relu_7	ReLU	5x13x256	
Conv_8	2D Convolution	5x13x256	256 3x3 convolutions with stride 1
batchnorm_9	Bach normalization	5x13x256	
Relu_8	ReLU	5x13x256	
Gap	2D gobal average pooling pooling	1 × 1 × 256	
Dropout	Dropout	1 × 1 × 256	50 %
Fc	Fully Connected	1 × 1 × 5	5 Fully Connected
Softmax	softmax	1 × 1 × 5	
Classoutput	Classification output	1 × 1 × 5	

convolution operation is given by (6):

$$Y_{m,n} = \sum_{j=1}^J \sum_{i=1}^I X_{m+i,n+j} * W_{ij} + b \quad (6)$$

Where X is the input image, and Y is the output. Its dimension is $M \times N$, with $1 \leq m \leq M$ and $1 \leq n \leq N$. W is the convolution kernel of size $I \times J$. It is in fact the matrix of weights which are updated during the training phase. b is the matrix of biases associated with weight.

C. Batch normalization layer

Batch normalization is a normalization operation performed between layers, rather than passing the raw data through the network. Its main role is to accelerate training by stabilizing the learning process. Batch normalization allows for the use of higher learning rates, which in turn facilitates faster convergence during training. The calculation for this operation is given by the following equation (7):

$$Z_N = \left(\frac{Z - M_Z}{S_Z} \right) \quad (7)$$

Where Z is the input. For this input, M_Z is the average, S_Z is the standard deviation. Z_N is the normalized output.

D. ReLU Layer

This layer applies an activation function: the Rectified Linear Unit (ReLU) function. It's a correction layer. It sets all negative values to zero and keeps the others.

E Fully connected layer

This layer returns at its output a vector of dimension N . Where N is the number of classes considered. It operates by performing a linear combination with weights and biases, followed by an activation function with the values received at the input.

F Softmax layer

This layer applies a particular activation function: the normalized exponential function. The Softmax layer converts an N -dimensional vector into a probability distribution over N classes. It is commonly used in neural networks for multi-class classification tasks. The Softmax function is applied at the output layer of the network to predict the probability of the input belonging each class. This function is defined by the following equation (8).

$$\sigma(z)_i = \frac{\exp(z_i)}{\sum_{j=1}^K \exp(z_j)} \quad (8)$$

With $1 \leq i \leq K$.

G. Classification output layer

This layer calculates the cross-entropy loss values for classification and weighted classification tasks with mutually exclusive classes. The cross-entropy loss measures how far the model's predictions are from the actual class labels. It works for both standard classification tasks and tasks where some classes are more important than others. It assumes that each sample belongs to only one class, meaning the classes do not overlap.

H Dropout layer

The principle of this layer is to randomly deactivate certain units of the network during training. It is a regularization technique that reduces the risk of overfitting by preventing neurons from over-specializing on the training data. Dropout helps the model to improve its ability to apply to new data. A dropout layer with 0.5 rate is used here.

3.4. Model optimization

Optimization of a deep learning model can be done in different ways. The goal is to improve its performance. The optimizer adjusts the model parameters during the training phase. In the proposed model, the Adam optimizer is used, which is commonly used. It is an improvement of the stochastic gradient descent (SGD) method, integrating additional techniques to enhance the convergence speed and handle different types of data. The learning rate is the hyperparameter that controls the step size used by the gradient descent algorithm during training in order to minimize the model's loss function. It controls the learning speed of a model.

K-fold cross-validation is a deep learning model evaluation technique that allows a better estimation of its performance thanks to a more efficient use of the available dataset. Its principle involves dividing the dataset into k-folds (subsets) of equal size, which are used to train and evaluate the model k times, each time using a different fold as a test set and the rest as a training set. The model's performance is then obtained by the average over the k tests. This technique provides a more reliable estimate of the model's performance.

The number of samples processed together in one pass through the network is called the batch size. Batch size influences the training efficiency. The batch processing optimizes the use of hardware resources and allows parallel computing. It allows to stabilize the training by smoothing the gradients. Finally, it influences the convergence speed of the model. Smaller batches allow more frequent updates of the model parameters, but with a lot of noise, while larger batches allow to have more stable gradients but with a less frequent update.

During the training of a deep learning model, a set of data is presented to it several times, adjusting each time the weights and biases of the network to minimize an error function. Each pass through the entire dataset is called an epoch, and each update to the model is an iteration.

Hyperparameters are obtained by optimizing the model via the generate and test method. The chosen hyperparameters are as follows: initial learning rate value: 0.00002; Batch size: 30; Number of epochs: 100.

3.5. Performance evaluation metrics

There are many performance evaluation metrics for classification tasks. The confusion matrix is a commonly used metric, which presents the predictions for each class. For each class, the confusion matrix contains the following components: the number of true positives (TP), the number of true negatives (TN), the number of false positives (FP), the number of false negatives (FN). From this matrix, other evaluation metrics can be derived, including specificity (Sp), precision (Pre), accuracy (Acc), sensitivity (Se) and F₁-score (F₁). These metrics are calculated as follows:

$$Pre = \frac{TP}{TP + FP} \quad (9)$$

$$Se = \frac{TP}{TP + FN} \quad (10)$$

$$Sp = \frac{TN}{TN + FP} \quad (11)$$

$$Acc = \frac{TP + TN}{TP + FP + TN + FN} \quad (12)$$

$$F1 = \frac{2 \cdot Se \cdot Pre}{Se + Pre} \quad (13)$$

4. Experimental results and discussion

The proposed method is experimented in the Matlab 2023b environment on a PC with an Intel Core i7 processor and 16 GB of RAM. The experiment has two phases such as training and testing. To ensure that the model performs well regardless of the dataset partitioning and to validate its classification performance, k-fold cross-validation is employed.

4.1. Experimental design and simulation procedure

i) Dataset Description

Two publicly available datasets is used for testing the proposed method.

- Bonn EEG Dataset:
 - o This dataset contains EEG signals recorded from 5 classes: A, B, C, D, and E.
 - o Each class consists of 100 instances (EEG segments).
 - o The EEG signals are sampled at 173.6 Hz, and each signal has a length of 4097 samples (approximately 23.6 s).
- NSC-D Dataset:
 - o This dataset contains EEG signals recorded from 3 classes: normal, interictal, and ictal.
 - o Each class contains 50 instances.
 - o The EEG signals are sampled at 200 Hz, with a signal length of 1024 samples. (approximately 5.12 s).

ii) Log-Mel Spectrogram Feature Extraction:

- o For each EEG window, the Mel spectrogram is computed by applying a Mel-filter bank.
- o Then a logarithmic transformation is applied to the Mel spectrogram to compress the dynamic range and highlight features of interest.

iii) Normalization:

- o The extracted Log-Mel spectrogram was normalized to zero mean and unit variance across each feature.

iv) Input Formatting:

- o These 2D images were used as input to the Convolutional Neural Network (CNN).

v) Model Architecture

A CNN is employed for the classification task.

vi) Experimental Procedure

The following procedure was followed for model training and testing.

- **Data Splitting:** The datasets were split into training and test sets with a train-test ratio of 7:3. This means that 70 % of the data was used for training the model, and 30 % was used for testing.
- **Cross-Validation:** the k-fold cross-validation is performed to evaluate the model's performance. The data was divided into k folds, and the model was trained and tested k times, each time using a different fold as the test set and the remaining folds as the training set.
- **Model Training:**
 - o The CNN was trained using the Adam optimizer with a learning rate of 0.00002;
 - o The training was performed for 100 epochs, and the model's performance was evaluated after each epoch.
- **Evaluation Metrics:** The model's performance was evaluated using the following metrics:
 - Accuracy (Acc)
 - Sensitivity (Se), which measures the model's ability to correctly identify seizure instances.
 - Specificity (Sp), which measures the model's ability to correctly identify non-seizure instances.
 - Precision (Pre), which measures the proportion of true positive predictions out of all positive predictions.
 - F₁-score (F₁), which is the harmonic mean of precision and recall.

4.2. Results on the bonn dataset

The proposed system is used to solve the following classification problems: A-E, B-E and AB-E (healthy and seizure), CD-E (pre-seizure and seizure), A-D-E, and AB-CD-E (healthy, pre-seizure and seizure). We also evaluate the five-class classification (A-B-C-D-E). From each class, 70 % of the dataset is used as the training set and the rest as the testing set. The training data is well separated from the test data. In fact, the network will be tested with data that was not used during training or validation. For the hyperparameters related

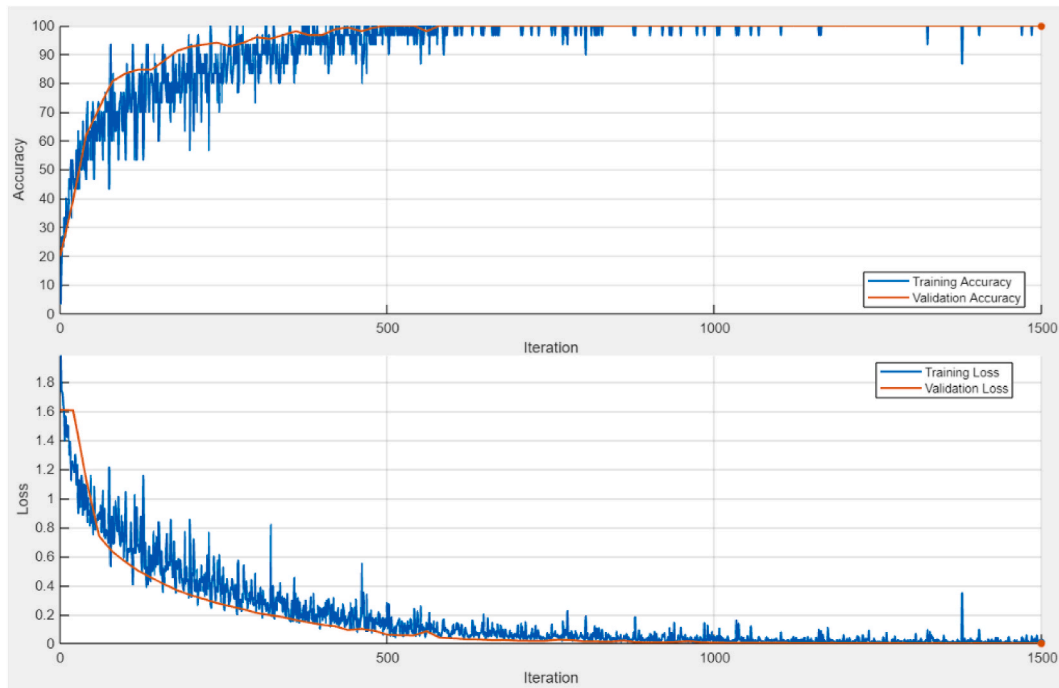


Fig. 5. Obtained training curves for the five-class classification on the Bonn Dataset.

to the Log-Mel Spectrogram, the Hamming window function $w[n]$ is adopted with a window length of $L = 102$, $M = 40$ Mel-filter banks, $H = 41$ for the hop size and a sampling rate $F_s = 173.6\text{Hz}$.

Fig. 5 presents results obtained during the training phase with the five-class classification. Table 2 presents confusion matrix obtained during testing phase. From this confusion matrix, the obtained average values are of 98.13 % for *Acc*, 95.33 % for *Se*, 98.83 % for *Sp*, 95.59 % for *Pre*, and 95.32 % for F_1 . It is important to note that the closer the value of each of these metrics is to 100 %, the better the result is.

Fig. 6 shows the training curves when considering tree classes AB-CD-E (healthy, pre-seizure and seizure). Table 3 presents the corresponding confusion matrix of the testing phase. The obtained average values are 99.56 %, 99.44 %, 99.63 %, 99.45 %, 99.44 % respectively for *Acc*, *Se*, *Sp*, *Pre*, and F_1 .

In order to validate the effectiveness of the proposed method, the k-fold cross-validation technique is used. Table 4 presents the obtained performance metrics with each of the 10 folds for the Bonn dataset, for the three-class classification. “std” is the standard deviation. As can be seen in this table, the average values of the performance metrics are $99.60\% \pm 00.64$, $99.33\% \pm 01.17$, $99.67\% \pm 00.54$, $99.52\% \pm 00.77$, $99.41\% \pm 00.98$, respectively for *Acc*, *Se*, *Sp*, *Pre*, and F_1 .

The results for the binary classification with 10-fold cross validation, on the Bonn dataset are shown in Table 5. For the A-E, B-E and AB-E classifications, we obtain $100\% \pm 00.00$ for all the performance metrics used. For CD-E, average values of $99.67\% \pm 00.01$, $100\% \pm 00.00$, $99.00\% \pm 03.16$, $99.52\% \pm 01.51$, $99.76\% \pm 00.77$ are obtained, respectively for *Acc*, *Se*, *Sp*, *Pre*, and F_1 .

4.3. Results on the NSC-ND dataset

The aim here is to distinguish between the three classes of EEG signals in this database. Two problems are considered: binary classification (normal-ictal) and triple classification (preictal-interictal-ictal). From each class, 70 % of the dataset is used as the training set and the rest as the testing set. For the LMS parameters, we use the Hamming window function for $w[n]$ with a window length $L = 28$, $M = 40$ Mel-filter banks, $H = 10$ for the hop size and $F_s = 200\text{ Hz}$ for the sampling rate. Since the number of EEG signal samples in this database is relatively small, we use 5-fold cross-validation to improve the reliability of the evaluation.

Fig. 7 presents obtained results during the training phase with the three-class classification on the NSC-ND dataset. Table 6 shows the test results. The performance metrics are $93.33\% \pm 03.14$, 90.00 ± 04.71 , 95.00 ± 02.36 , 91.89 ± 03.30 , 89.50 ± 05.44 , respectively for *Acc*, *Se*, *Sp*, *Pre*, and F_1 .

Table 7 shows the obtained test result for the binary classification (Normal-Ictal). As can be seen, the performance metric values are $98.67\% \pm 01.83$, 98.00 ± 02.74 , 100 ± 00.00 , 100 ± 00.00 , 98.97 ± 01.40 , respectively for *Acc*, *Se*, *Sp*, *Pre*, and F_1 .

In order to evaluate the robustness of the proposed method and facilitate an effective comparison with other state-of-the-art techniques, we varied the training-to-test ratios (TTTR). The TTTR refers to the proportion of data used for training a model compared to the data used for testing. This ratio is a crucial factor in performance evaluation as it directly affects the model’s ability to generalize. Table 8 presents the results obtained with different k-fold cross validation. As shown in Table 8, for the classification tasks A-E, AB-E, and CD-E, the performance remains unchanged across the distinct TTTRs considered. For the other tasks, the impact of the TTTR on model performance is minimal. However, a slightly more pronounced difference is observed for the NSC-ND dataset. This reflects the fact suggests that deep learning models require a sufficient amount of data for training, and that a training of only 5 samples (1:9 split from 50 samples) in the NSC-ND database is insufficient for good generalization.

4.4. Discussion

In this study, we proposed a CNN-based method using log-Mel spectrograms to classify EEG signals, specifically for the detection of epileptic seizures. The performances of the proposed model are evaluated on two datasets: the Bonn and the NSC-ND datasets. These are public, free and available EEG signal database generally used for this purpose. Table 9 presents the obtained performance metrics compared with other methods in the literature on the same databases. The best results are shown in bold. This experimental result demonstrates the effectiveness and robustness of the proposed approach, showing significant improvements in classification accuracy, sensitivity, specificity, and precision compared to several state-of-the-art (SOTA) methods. In this section, we provide an in-depth discussion of the results and their implications for both the academic community and practical applications.

Table 2
Confusion matrix and performance metrics for the five-class classification on the Bonn dataset.

Confusion matrix and performance metrics for the five class classification on the 2016 dataset											
	Predicted Class					Performance metrics					
		A	B	C	D	E	Acc (%)	Se (%)	Sp (%)	Pre (%)	F ₁ (%)
True class	A	30	0	0	0	0	98.67	100	98.33	93.75	96.77
	B	0	29	0	1	0	96.67	96.67	96.67	87.88	92.06
	C	2	0	28	0	0	98.67	93.33	100	100	96.65
	D	0	4	0	26	0	96.67	86.67	99.17	96.30	91.23
	E	0	0	0	0	30	100	100	100	100	100
Average							98.13	95.33	98.83	95.59	95.32

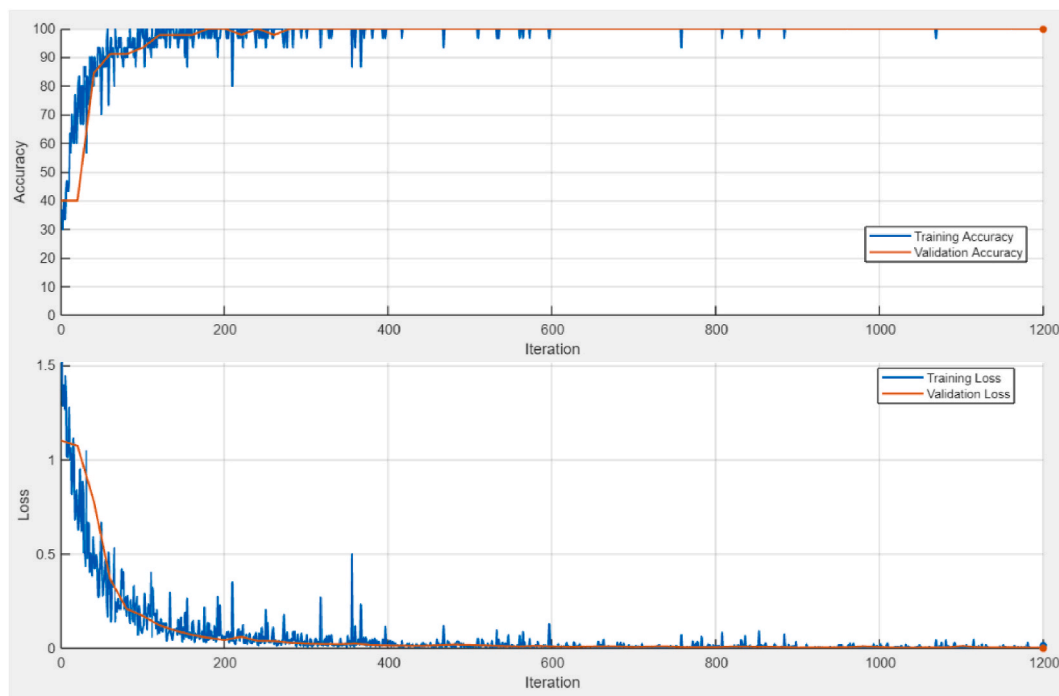


Fig. 6. Obtained training curves for the three-class classification on the Bonn Dataset.

Table 3

Confusion matrix and performance metrics for three-class classification on the Bonn dataset.

	Predicted Class				Performance metrics				
		AB	CD	E	Acc (%)	Se (%)	Sp (%)	Pre (%)	F ₁ (%)
True class	AB	60	0	0	99.33	100	98.89	98.36	99.17
	CD	1	59	0	99.33	98.33	100	100	99.16
	E	0	0	30	100	100	100	100	100
Average					99.56	99.44	99.63	99.45	99.44

Table 4

Performance metrics for the three-class classification on the Bonn dataset with 10-fold.

# fold	1	2	3	4	5	6	7	8	9	10	Mean	std
Acc (%)	100	100	100	100	98.67	98.67	98.67	100	100	100	99.60	00.64
Se (%)	100	100	100	100	96.67	98.33	93.33	100	100	100	99.33	01.17
Sp (%)	100	100	100	100	98.89	98.89	98.89	100	100	100	99.67	00.54
Pre (%)	100	100	100	100	98.41	98.41	98.41	100	100	100	99.52	00.77
F ₁ (%)	100	100	100	100	97.43	98.33	98.33	100	100	100	99.41	00.98

4.4.1. Performance comparison with existing methods

The proposed approach consistently outperforms existing methods, particularly when compared to those using 1D CNNs on raw EEG signals. For example, Acharya et al. (2018) [38] used a deep CNN with 13 layers to detect seizures on the Bonn dataset, achieving an accuracy of 88.67 % with 95 % sensitivity, 90 % specificity, and 99.51 % precision, for the AB-CD-E classification task. The proposed method, however, surpasses these results with 99.60 % accuracy, 99.17 % sensitivity, 99.67 % specificity, and 99.51 % precision, even with a more challenging 4:1 train-test ratio. This superior performance demonstrates the advantages of using log-Mel spectrograms, which effectively capture time-frequency information from EEG signals, as compared to raw signals or hand-crafted features.

In comparison to the work by Mallick et al. (2024) [49], who used a combination of 1D CNNs, Bidirectional LSTMs, and GRU, the proposed method performs slightly lower on certain tasks (e.g., A-B-C-D-E), but outperforms them in the AB-CD-E classification task. Mandhouj et al. (2021) [19] also used time-frequency representations like short-time Fourier transform (STFT) spectrograms with CNNs and achieved near-perfect results for specific class combinations. However, the proposed model maintains higher accuracy and sensitivity for more complex combinations, such as AB-CD-E, showing the robustness and generalization ability of the proposed

Table 5
Performance metrics for the binary classification on the Bonn Dataset.

	# fold	1	2	3	4	5	6	7	8	9	10	Mean	std
A-E	Acc (%)	100	100	100	100	100	100	100	100	100	100	100	00.00
	Se (%)	100	100	100	100	100	100	100	100	100	100	100	00.00
	Sp (%)	100	100	100	100	100	100	100	100	100	100	100	00.00
	Pre (%)	100	100	100	100	100	100	100	100	100	100	100	00.00
	F ₁ (%)	100	100	100	100	100	100	100	100	100	100	100	00.00
B-E	Acc (%)	100	100	100	100	100	100	100	100	100	100	100	00.00
	Se (%)	100	100	100	100	100	100	100	100	100	100	100	00.00
	Sp (%)	100	100	100	100	100	100	100	100	100	100	100	00.00
	Pre (%)	100	100	100	100	100	100	100	100	100	100	100	00.00
	F ₁ (%)	100	100	100	100	100	100	100	100	100	100	100	00.00
AB-E	Acc (%)	100	100	100	100	100	100	100	100	100	100	100	00.00
	Se (%)	100	100	100	100	100	100	100	100	100	100	100	00.00
	Sp (%)	100	100	100	100	100	100	100	100	100	100	100	00.00
	Pre (%)	100	100	100	100	100	100	100	100	100	100	100	00.00
	F ₁ (%)	100	100	100	100	100	100	100	100	100	100	100	00.00
CD-E	Acc (%)	100	100	100	100	100	100	100	100	96.67	100	99.67	00.01
	Se (%)	100	100	100	100	100	100	100	100	100	100	100	00.00
	Sp (%)	100	100	100	100	100	100	100	100	90.00	100	99.00	03.16
	Pre (%)	100	100	100	100	100	100	100	100	95.24	100	99.52	01.51
	F ₁ (%)	100	100	100	100	100	100	100	100	97.56	100	99.76	00.77

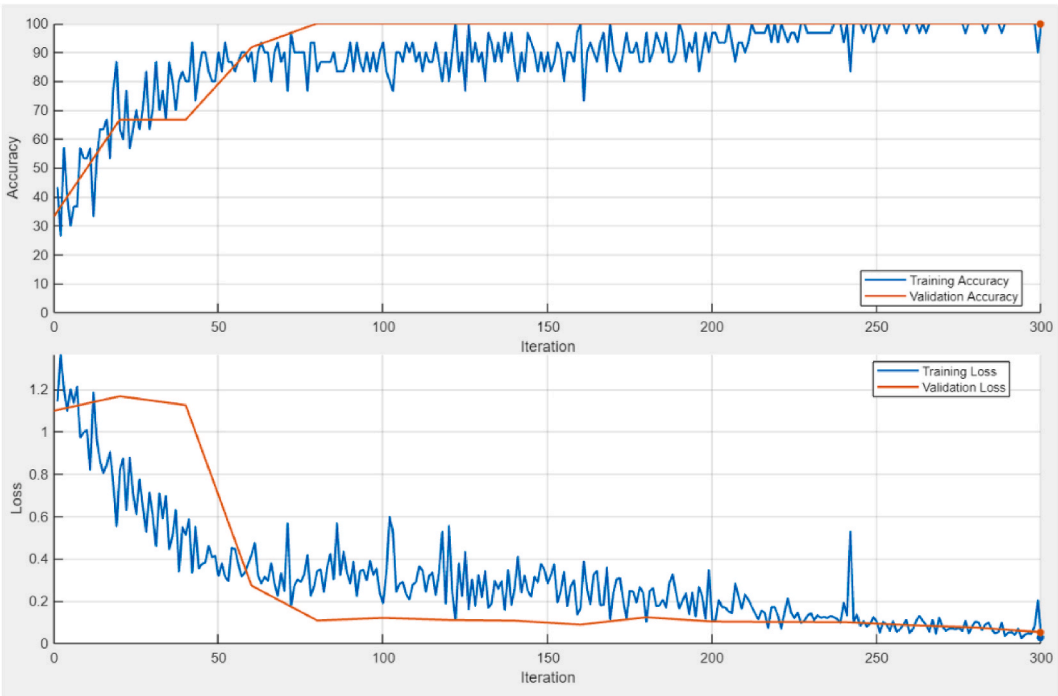


Fig. 7. Obtained training curves for the three-class classification on the NSC-ND dataset.

Table 6
Performance metrics for the triple classification on the NSC-ND dataset with 5-fold.

# fold	1	2	3	4	5	Mean	Std
Acc (%)	91.11	95.56	88.89	95.56	95.56	93.33	03.14
Se (%)	86.67	93.33	83.33	93.33	93.33	90.00	04.71
Sp (%)	93.33	96.67	91.67	96.67	96.67	95.00	02.36
Pre (%)	90.48	94.44	86.75	93.33	94.44	91.89	03.30
F ₁ (%)	86.11	93.27	81.51	93.33	93.27	89.50	05.44

Table 7

Performance metrics for the binary classification on the NSC-ND dataset with 5-fold.

# fold	1	2	3	4	5	Mean	Std
Acc (%)	100	100	96.67	100	96.67	98.67	01.83
Se (%)	100	100	95.00	100	95.00	98.00	02.74
Sp (%)	100	100	100	100	100	100	00.00
Pre (%)	100	100	100	100	100	100	00.00
F ₁ (%)	100	100	97.44	100	97.44	98.97	01.40

Table 8

Performance variation as a function of training-to-test ratio.

Dataset	Classes	TTR	Acc (%)	Se (%)	Sp (%)	Pre (%)	F ₁ (%)
Bonn	A-E	1:1	100	100	100	100	100
		4:1	100	100	100	100	100
		9:1	100	100	100	100	100
	B-E	4:1	99.50 ± 01.12	100	99.00 ± 2.24	99.05 ± 2.13	99.51 ± 1.09
		9:1	100	100	100	100	100
	AB-E	4:1	100	100	100	100	100
		9:1	100	100	100	100	100
	CD-E	4:1	99.67 ± 00.75	100	99.00 ± 2.24	99.52 ± 1.09	99.76 ± 0.55
		9:1	99.67 ± 00.01	100	99.00 ± 03.16	99.52 ± 01.51	99.76 ± 00.77
	AB-CD-E	4:1	99.60 ± 00.60	99.17 ± 01.18	99.67 ± 00.50	99.51 ± 00.74	99.33 ± 00.97
		9:1	99.60 ± 00.64	99.33 ± 01.17	99.67 ± 00.54	99.52 ± 00.77	99.41 ± 00.98
	A-D-E	4:1	99.33 ± 00.99	99.00 ± 01.49	99.50 ± 00.75	99.08 ± 01.36	99.00 ± 01.49
		9:1	99.56 ± 00.94	99.33 ± 01.41	99.67 ± 00.70	99.39 ± 01.28	99.33 ± 01.41
		1:1	99.56 ± 00.00	99.33 ± 00.00	99.67 ± 00.00	99.35 ± 00.00	99.33 ± 00.00
NSC-ND	Normal-Ictal	9:1	100	100	100	100	100
		1:9	97.63 ± 00.98	96.44 ± 01.46	100	100	98.18 ± 00.76
	Preictal-Interictal-Ictal	9:1	93.78 ± 04.29	90.67 ± 06.44	95.33 ± 03.22	92.63 ± 05.03	90.19 ± 07.1
		1:9	92.4 ± 00.66	88.67 ± 00.99	94.33 ± 00.50	89.91 ± 00.95	88.32 ± 01.04

approach.

Additionally, Turk et al. (2019) [52] employed a CNN combined with scalogram features, achieving 99.50 % accuracy for the A-E task and 97.36 % for the A-B-C-D-E classification. The proposed method achieves 100 % accuracy for binary classifications and 98.13 % for the multi-class A-B-C-D-E task, suggesting that log-Mel spectrograms may provide a more compact and informative time-frequency representation than scalograms for certain EEG signals. This observation is further validated when comparing obtained results to those of Varli et al. (2023) [48], who used continuous wavelet transform (CWT) and spectrogram (STFT) with deep learning, where the proposed model outperforms them in all tasks.

4.4.2. Advantages of log-Mel spectrograms and CNNs

The key advantage of the proposed approach lies in the log-Mel spectrogram feature extraction process combined with CNNs. The log-Mel spectrograms used in the proposed approach provide a compact yet rich time-frequency representation of EEG signals. This method provides a straightforward yet effective representation of EEG data, which is both interpretable and computationally efficient. For example, Kaur et al. (2023) [50] achieved high performance using deep features extracted from AlexNet and EEG scalograms, but the proposed model outperforms theirs in all tested classification tasks, including more challenging splits such as AB-CD-E.

Furthermore, the proposed method also outperforms traditional time-frequency analysis techniques, such as those used by Tzallas et al. (2009) [5], who employed Cohen's class time-frequency distributions with an ANN, achieving 100 % accuracy for the A-E and A-D-E tasks. While Tzallas et al. achieved strong results for these simpler tasks, the proposed method maintains a higher level of accuracy (98.13 %) on the A-B-C-D-E classification task. Similarly, Acharya et al. (2013) [4] used wavelet transforms combined with ICA features and traditional classifiers, achieving 96 % accuracy. The proposed approach improves on this, achieving 99.11 % accuracy in the A-D-E classification task.

The log-Mel spectrogram's ability to distill the essential features of the EEG signal into a compact form allows the CNN to more effectively distinguish between classes, particularly in the context of multi-class seizures where traditional methods may struggle. This advantage is particularly evident when comparing the proposed method to recent works like Kaur et al. (2023) and Varli et al. (2023), where the proposed model consistently achieves higher performance across both simple and challenging classification tasks.

4.4.3. Performance under varying training-test ratios

One of the key strengths of the proposed method is its robustness across various datasets and class combinations. Even when trained with limited data, the proposed model maintains high performance. This is particularly evident in the 1:9 train-test ratio scenario for the Normal-Ictal classification task on the NSC-ND dataset, where we achieved 97.63 % accuracy, 96.44 % sensitivity, and 100 % specificity and precision, outperforming several traditional methods. For instance, Swami et al. (2017, 2019) [62,63] used dual-tree complex wavelet transform (DTCWT) combined with generalized regression neural networks (GRNN) on the NSC-ND dataset,

Table 9

Performance metrics compared with other state-of-the-art methods. Bold values indicate the best results. “-” refers to not available.

Authors	Methods		Dataset	Classes	TTTR	Acc (%)	Se (%)	Sp (%)	Pre (%)	F ₁ (%)
Tzallas et al. (2009) [5]	Time frequency analysis + ANN		Bonn	A-E A-B-C-D-E A-D-E	1:1	100 89.00 100	- - -	- - -	- - -	- - -
Gandhi et al. (2012) [61]	DWT + PNN + DHS MD		NSC-ND	Normal-Ictal	9:1	100	100	100	-	-
Acharya et al. (2013) [4]	DWT, ICA coefficients + SVM		Bonn	A-D-E	9:1	96	96	97	98.6	-
Swami et al., 2016 [67]	Mean DTCWT + GRNN		Bonn	A-E B-E AB-E CD-E Normal-Ictal	9:1	100 100 100 99.67 99.67	100 100 100 100 100	100 100 100 100 99.50	- - - - -	- - - - -
Swami et al., 2017 [63]	DT-CWT-ENE-STDF + GRNN		NSC-ND	Normal-Ictal	1:9	98.24	98.34	98.35	-	-
Zahra et al. (2017) [37]	Multivariate Empirical Mode Decomposition + ANN		Bonn	A-B-C-D-E	6.5:3.5	-	87.2	-	-	-
Tiwari et al. (2017) [33]	Key point-based LBP + SVM		Bonn	AB-E CD-E AB-CD-E	9:1	100 99.45 98.80	100 99.68 -	100 99.00 -	100 99.42 -	- - -
Acharya et al. (2018) [38]	1D-CNN		Bonn	AB-CD-E	7:3	88.67	95.00	90.00	95.00	-
Swami et al., 2019 [62]	DT-CWT + GRNN		NSC-ND	Normal-Ictal	1:9	98.12	97.49	98.74	-	-
Subasi et al. (2019) [21]	PSO + SVM		Bonn	A-E	7.5:2.5	99.38	99.50	99.25	-	-
Turk et al. (2019) [52]	CNN + Scalogram		Bonn	A-E B-E A-B-C-D-E	9:1	99.50 99.50 97.36	99.00 100 93.6	100 100 98.33	- - -	99.5 99.50 93.58
Raghu et al. (2019) [16]	Matrix determinant + MLP		Bonn	A-E B-E AB-E CD-E AB-CD-E	9:1	99.45 96.06 97.10 96.95 94.35	99.70 96.10 97.50 97.60 96.40	99.20 92.95 98.50 94.30 92.30	99.25 92.12 98.75 94.25 96.20	- - - - -
Gupta et al. (2019) [34]	FBSE + WMRPE + LS-SVM		Bonn	A-E B-E C-E D-E CD-E	9:1	99.50 99.50 99.50 97.50 99.00	- - - - -	- - - - -	- - - - -	- - - - -
Carvalho et al. (2019) [65]	Adaptive decomposition methods + MLP, KNN Linear SVM, RBF SVM, GPC	EMD EEMD CEEMDAN EWT VMD Original signal	Bonn	A-D-E	9:1	97.7 98.6 98.5 97.1 98.3 95.1	96.4 96.0 97.8 95.9 97.9 93.0	98.4 99.9 99.4 97.7 98.6 96.2	- - - - - -	- - - - - -
		EMD EEMD CEEMDAN EWT VMD Original signal	NSC-ND	Normal-Ictal		99.0 100 98.4 100 100 98.1	98.0 100 96.8 100 100 96.6	100 100 100 100 100 99.6	- - - - - -	- - - - - -
		EMD EEMD CEEMDAN EWT VMD		Preictal-interictal-ictal		[82.2, 90.4] [79.9, 85.3] [78.5, 93.1] [78.2, 91.3] [88.3, 92.1]	- - - - -	- - - - -	- - - - -	

(continued on next page)

Table 9 (continued)

Authors	Methods	Dataset	Classes	TTTR	Acc (%)	Se (%)	Sp (%)	Pre (%)	F ₁ (%)
			Original signal		[71.3, 86.7]	–	–	–	–
Akyol (2020) [41]	Stacking ensemble based deep learning approach	Bonn	A-E	9:1	97.17	93.11	98.18	–	–
Abiyev et al. (2020) [42]	1D-CNN	Bonn	AB-CD-E	9:1	98.67	97.67	98.83	–	–
Mandhouj et al. (2021) [19]	STFT Spectrogram with deep CNN	Bonn	A-E	7:3	100	100	100	100	–
			B-E		98.8	99.17	99.17	98.39	–
			AB-E		98.8	99.17	99.17	98.39	–
			CD-E		98.8	99.17	99.17	98.39	–
			AB-CD-E		98.22	97.78	98.61	97.31	–
Kaur et al. (2023) [50]	Deep features from the AlexNet using EEG scalogram + SVM	Bonn	A-E	9:1	98.64	–	–	–	–
			B-E		100	–	–	–	–
			AB-E		100	–	–	–	–
			AB-CD-E		98.98	–	–	–	–
		NSC-ND	Normal-Ictal		98.67	100	96.00	–	–
Varli and Yilmaz (2023) [48]	Combined deep learning using CWT	Bonn	A-E	7:1	98.97	98.97	98.97	98.98	98.97
			B-E		99.62	99.2	99.62	99.63	99.62
			AB-E		99.46	99.46	99.26	99.46	99.46
			CD-E		98.92	98.92	98.55	98.92	98.92
			AB-CD-E		97.30	97.30	98.35	97.31	97.30
	Combined deep learning using STFT		A-E		99.81	99.81	99.81	99.81	99.81
			B-E		99.94	99.94	99.94	99.94	99.94
			AB-E		99.81	99.81	99.81	99.81	99.81
			CD-E		99.17	99.17	99.05	99.17	99.17
			AB-CD-E		98.20	98.20	98.99	98.20	98.09
Mallick et al. (2024) [49]	1D-CNNs, Bidirectional LSTMs, GRU	Bonn	A-E	4:1	100	100	100	–	–
			B-E		100	100	100	–	–
			AB-CD-E		99.2	99.14	99.55	–	–
			A-B-C-D-E		98.18	98	99.49	–	–
Proposed Method (2024)	log-Mel Spectrogram + CNN	Bonn	A-E	9:1	100	100	100	100	100
			B-E		100	100	100	100	100
			AB-E		100	100	100	100	100
			CD-E		99.67	100	99.00	99.52	99.76
			AB-CD-E		99.60	99.33	99.67	99.52	99.41
			A-D-E		99.11	98.67	99.33	98.73	98.67
			A-E	4:1	100	100	100	100	100
			B-E		99.50	100	99.00	99.05	99.51
			AB-E		100	100	100	100	100
			CD-E		99.67	100	99.00	99.52	99.76
			AB-CD-E		99.60	99.17	99.67	99.51	99.33
			A-D-E		99.33	99.00	99.50	99.08	99.00
			A-D-E	1:1	99.56	99.33	99.67	99.35	99.33
			A-B-C-D-E	7:3	98.13	95.33	98.83	95.59	95.32
		NSC-ND	Normal-Ictal	9:1	100	100	100	100	100
			Preictal-Ictal		93.33	90.00	95.00	91.89	89.50
			Normal-Ictal	1:9	97.63	96.44	100	100	98.18
			Preictal-Ictal		92.4	88.67	94.33	89.91	88.32
			Normal-Ictal	4:1	98.67	98.00	100	100	98.97
			Preictal-Ictal		93.33	90.00	95.00	91.89	89.50

achieving high sensitivity and specificity values around 98 %. However, the proposed model excels in maintaining high specificity and precision.

4.4.4. Practical applications and real-world integration

The proposed method shows great potential for real-world applications, particularly in embedded systems for medical diagnostics. With high classification accuracy and fast training times, the proposed model can be integrated into devices for real-time EEG signal classification, aiding in the diagnosis of epileptic seizures. Additionally, the log-Mel spectrogram feature extraction process is simple and effective, which makes the model both efficient and interpretable—a critical factor for clinical adoption.

4.4.5. Limitations and future directions

While the proposed method demonstrates promising results, there are several limitations. First, the model has been evaluated on specific datasets (Bonn and NSC-ND), and its generalizability to other epilepsy datasets or different seizure types requires further validation. Additionally, the method relies on high-quality EEG data, which may not always be available in clinical settings. While the computational complexity is reasonable for modern systems, it may still pose challenges for real-time applications in resource-constrained environments.

Future research could focus on expanding the generalization of the model by testing it on a wider variety of EEG datasets, including those from different geographical regions or involving patients with rare seizure types.

4.4.6. Broader impact

The broader impact of this research extends beyond epilepsy diagnosis. The deep learning methods and feature extraction techniques presented could be applied to other neurological disorders, such as Parkinson's disease, Alzheimer's disease, and sleep disorders, where EEG analysis plays a crucial role in diagnosis and monitoring. By enabling more accurate and timely seizure detection, the proposed approach can enhance the quality of life for individuals with epilepsy, reduce the burden on healthcare professionals, and contribute to the development of wearable, real-time seizure monitoring systems.

5. Conclusion

The method proposed in this paper uses the EEG signal features extracted from the logarithm of the spectrogram in the Mel scale. These features are used to train a convolutional neural network. For the experiment, we tested the proposed method on the classification of EEG signals from the Bonn and NSC-ND databases, for the diagnosis of epileptic seizures. These public databases contain many classes of EEG signals each characterizing a brain state, either the presence or absence of the epileptic seizure or the moment preceding the seizure. We performed k-fold cross-validation to show if the proposed model could obtain successful results across entire datasets. The obtained results are quite interesting compared to those of other state-of-the-art methods. The model excels in both binary and multi-class classification tasks, outperforming many existing methods in terms of accuracy, sensitivity, specificity, and F₁-score. This result justifies the effectiveness of the use of the log Mel spectrogram as features in the classification of EEG signals and particularly for the diagnosis of epileptic seizures. The robustness of the model across different TTR settings and its impressive performance on the NSC-ND dataset further underline its potential for practical applications in epilepsy diagnosis. We believe that the simplicity, efficiency, and accuracy of the proposed method make it a strong candidate for deployment in real-world medical systems, paving the way for faster, more accurate seizure detection in clinical practice. Future work will focus on expanding the model's applicability by integrating more diverse datasets and exploring methods to improve robustness against noisy data.

CRediT authorship contribution statement

Beaudelaire Saha Tchinda: Conceptualization, Methodology, Software, Validation, Visualization, Writing – original draft, Writing – review & editing. **Daniel Tchiotso:** Supervision, Validation, Visualization, Writing – review & editing. **Laurent Chanel Djoufack Nkengfack:** Validation, Visualization, Writing – review & editing. **René Tchinda:** Supervision, Visualization, Writing – review & editing.

Consent to participate

Not applicable.

Consent to publish

Not applicable.

Data availability statement

Datasets generated during the current study are available from the corresponding author on reasonable request.

Ethics approval

This article does not contain any studies with human participants or animals performed by any of the authors.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

- [1] Organisation mondiale de la Santé, Épilepsie. <https://www.who.int/fr/news-room/fact-sheets/detail/epilepsy>, 2024.
- [2] R.S. Fisher, W. van Emde Boas, W. Blume, C. Elger, P. Genton, P. Lee, J. Engel Jr., Epileptic seizures and epilepsy: definitions proposed by the international league against epilepsy (ILAE) and the international bureau for epilepsy (IBE), *Epilepsia* (46) (2005) 470–472.
- [3] X.L. Ma, S. Qiu, W. Wei, S.P. Wang, H.G. He, Deep Channel-correlation network for motor imagery decoding from the same limb, *IEEE Trans. Neural Syst. Rehabil. Eng.* 28 (1) (2019) 297–306, <https://doi.org/10.1109/TNSRE.2019.2953121>.
- [4] Ur Acharya, R. Yanti, G. Swapna, Vs Sree, Rj Martis, Js Suri, Automated diagnosis of epileptic electroencephalogram using independent component analysis and discrete wavelet transform for different electroencephalogram durations, *Proc. IME H J. Eng. Med.* 227 (2013) 234–244, <https://doi.org/10.1177/0954411912467883>.
- [5] A.T. Tzallas, M.G. Tsipouras, D.I. Fotiadis, Epileptic seizure detection in EEGs using time–frequency analysis, *IEEE Trans. Inf. Technol. Biomed.* 13 (2009) 703–710, <https://doi.org/10.1109/TTB.2009.2017939>.
- [6] Z.K. Gao, Y.L. Li, Y.X. Yang, X.M. Wang, N. Dong, H.D. Chiang, A GPSO-optimized convolutional neural networks for EEG based emotion recognition, *Neurocomputing* 380 (2020) 225–235, <https://doi.org/10.1016/j.neucom.2019.10.096>.
- [7] J.D. Tao, Z. Yin, L. Liu, Y. Tian, Z.Q. Sun, J.H. Zhang, Individual-specific classification of mental workload levels via an ensemble heterogeneous extreme learning machine for EEG modeling, *Symmetry-Basel* 11 (2019) 944, <https://doi.org/10.3390/SYM11070944>.
- [8] P.M. Kulkarni, Z.D. Xiao, E.J. Robinson, A.S. Jami, J.P. Zhang, H.C. Zhou, S.E. Henin, A.A. Liu, R.S. Osorio, J. Wang, Z. Chen, A deep learning approach for real-time detection of sleep spindles, *J. Neural. Eng.* 16 (3) (2019), <https://doi.org/10.1088/1741-2552/2Fab0933>.
- [9] O. Ozdenizci, Y. Wang, T. Koike-Akino, D. Erdogmus, Adversarial deep learning in EEG biometrics, *IEEE Signal Process. Lett.* 26 (5) (2019) 710–714, <https://doi.org/10.1109/LSP.2019.2906826>.
- [10] P. Kaushik, A. Gupta, P.P. Roy, D.P. Dogra, EEG-based age and gender prediction using deep BLSTM-LSTM network model, *IEEE Sens. J.* 19 (7) (2019) 2634–2641, <https://doi.org/10.1109/JSEN.2018.2885582>.
- [11] Jyoti Pillai, R. Michael Sperling, Interictal EEG and the diagnosis of epilepsy, *Epilepsia* 47 (Suppl 1) (2006) 14–22, <https://doi.org/10.1111/j.1528-1167.2006.00654.x>. ISSN 0013-580. PMID 17044820. S2CID 8668713.
- [12] S. Raghu, N. Sriraam, Y. Temel, S. Vasudeva Rao, P.L. Kubben, EEG based multi-class seizure type classification using convolutional neural network and transfer learning, *Neural Netw.* 124 (2020) 202–212, <https://doi.org/10.1016/j.neunet.2020.01.017>. ISSN 0893-6080.
- [13] S. Roy, U. Asif, J. Tang, S. Harrer, Seizure type classification using EEG signals and machine learning: setting a benchmark. 2020 IEEE Signal Processing in Medicine and Biology Symposium (SPMB), 2020, pp. 1–6, <https://doi.org/10.1109/SPMB50085.2020.9353642>. Philadelphia, PA, USA.
- [14] S. Gong, K. Xing, A. Cichocki, J. Li, Deep learning in EEG: advance of the last ten-year critical period, *IEEE Transactions on Cognitive and Developmental Systems* (2021) 1–39, <https://doi.org/10.1109/TCDS.2021.3079712>.
- [15] S. Aydin, H.M. Saraoglu, S. Kara, Log energy entropy-based EEG classification with multilayer neural networks in seizure, *Ann. Biomed. Eng.* 37 (2009) 2626–2630, <https://doi.org/10.1007/s10439-009-9795-x>.
- [16] S. Sriraam, N. Hegde, A.S. Kubben, A novel approach for classification of epileptic seizures using matrix determinant, *Expert Syst. Appl.* 127 (1) (2019) 323–341, <https://doi.org/10.1016/j.eswa.2019.03.021>.
- [17] V. Gupta, A. Nishad, R.B. Pachori, Focal EEG signal detection based on constant-bandwidth TQWT filter-banks. 2018 IEEE International Conference on Bioinformatics and Biomedicine (BIBM), 2018, pp. 2597–2604, <https://doi.org/10.1109/BIBM.2018.8621311>. Madrid, Spain.
- [18] S. Raghu, N. Sriraam, G.P. Kumar, Classification of epileptic seizures using wavelet packet log energy and norm entropies with recurrent Elman neural network classifier, *Cogn Neurodyn* 11 (2017) 51–66, <https://doi.org/10.1007/s11571-016-9408-y>.
- [19] B. Mandhouj, M.A. Cherni, M. Sayadi, An automated classification of EEG signals based on spectrogram and CNN for epilepsy diagnosis, *Analog Integr Circ Sig Process* 108 (2021) 101–110, <https://doi.org/10.1007/s10470-021-01805-2>.
- [20] G. Chen, W. Xie, T.D. Bui, A. Krzyzak, Automatic epileptic seizure detection in EEG using non subsampled wavelet–fourier features, *J. Med. Biol. Eng.* 37 (1) (2017) 123–131, <https://doi.org/10.1007/s40846-016-0214-0>.
- [21] A. Subasi, J. Kevric, M. Abdullah Canbaz, Epileptic seizure detection using hybrid machine learning methods, *Neural Comput & Applic* 31 (2019) 317–325, <https://doi.org/10.1007/s00521-017-3003-y>.
- [22] T. Zhang, W. Chen, LMD based features for the automatic seizure detection of EEG signals using SVM, *IEEE Trans. Neural Syst. Rehabil. Eng.* 25 (8) (Aug. 2017) 1100–1108, <https://doi.org/10.1109/TNSRE.2016.2611601>.
- [23] Y. Jiang, et al., Seizure classification from EEG signals using transfer learning, semi-supervised learning and TSK fuzzy system, *IEEE Trans. Neural Syst. Rehabil. Eng.* 25 (12) (Dec. 2017) 2270–2284, <https://doi.org/10.1109/TNSRE.2017.2748388>.
- [24] S. Raghu, N. Sriraam, S.V. Rao, A.S. Hegde, P.L. Kubben, Automated detection of epileptic seizures using successive decomposition index and support vector machine classifier in long-term EEG, *Neural Comput & Applic* 32 (2020) 8965–8984, <https://doi.org/10.1007/s00521-019-04389-1>.
- [25] S. Raghu, Natarajan Sriraam, Yasin Temel, Shyam Vasudeva Rao, Alangar Satyanandadas Hegde, Pieter L. Kubben, Performance evaluation of DWT based sigmoid entropy in time and frequency domains for automated detection of epileptic seizures using SVM classifier, *Comput. Biol. Med.* 110 (2019) 127–143, <https://doi.org/10.1016/j.combiomed.2019.05.016>. ISSN 0010-4825.
- [26] M. Mumtaz, M. Afzal, A. Mushtaq, Sensorimotor cortex EEG signal classification using hidden Markov models and wavelet decomposition. 2018 IEEE International Symposium on Signal Processing and Information Technology (ISSPIT), 2018, pp. 375–381, <https://doi.org/10.1109/ISSPIT.2018.8642672>. Louisville, KY, USA.
- [27] L.C. Djoufack Nkengfack, D. Tchiotso, R. Atangana, B. Saha Tchinda, V. Louis-Door, D. Wolf, Jacobi polynomial transforms-based entropy measures for focal and non-focal EEG signals discrimination using kernel machines, *Sci. J. Circ. Syst. Signal Process.* 10 (No.2) (2021) 25–37, <https://doi.org/10.11648/j.cssp.20211002.11>, 2021.
- [28] H. Chiang, M. Chen, Y. Huang, Wavelet-based EEG processing for epilepsy detection using fuzzy entropy and associative Petri net, *IEEE Access* 7 (2019) 103255–103262, <https://doi.org/10.1109/ACCESS.2019.2929266>.
- [29] B. Peachap Atemangho, D. Tchiotso, Epileptic seizures detection based on some new Laguerre polynomial wavelets, artificial neural networks and support vector machines, *Inform. Med. Unlocked* 16 (2019), <https://doi.org/10.1016/j.imu.2019.100209>, 2019.
- [30] L.C. Djoufack Nkengfack, D. Tchiotso, R. Atangana, V. Louis-Door, D. Wolf, EEG signals analysis for epileptic seizures detection using polynomial transforms, linear discriminant analysis and support vector machines, *Biomed. Signal Process Control* 62 (2020) 102141, <https://doi.org/10.1016/j.bspc.2020.102141>.
- [31] L.C. Djoufack Nkengfack, D. Tchiotso, R. Atangana, B. Saha Tchinda, Valérie Louis-Door, Didier Wolf, A comparison study of polynomial-based PCA, KPCA, LDA and GDA feature extraction methods for epileptic and eye states EEG signals detection using kernel machines, *Inform. Med. Unlocked* 26 (2021) 100721, <https://doi.org/10.1016/j.imu.2021.100721>.
- [32] N.J. Sairamya, M.S.P. Subathra, S. Easter Selvan, S. Thomas George, A new approach for automatic detection of focal EEG signals using wavelet packet decomposition and quad binary pattern method, *Biomed. Signal Process Control* 63 (2021) 102096, <https://doi.org/10.1016/j.bspc.2020.102096>. ISSN 1746-8094.

- [33] A.K. Tiwari, R.B. Pachori, V. Kanhangad, B.K. Panigrahi, Automated diagnosis of epilepsy using key-point-based local binary pattern of EEG signals, *IEEE J. Biomed. Health Inform.* 21 (2017) 888–896, <https://doi.org/10.1109/JBHI.2016.25.89971>.
- [34] V. Gupta, R.B. Pachori, Epileptic seizure identification using entropy of FBSE based EEG rhythms, *Biomed. Signal Process Control* 53 (2019) 101569, <https://doi.org/10.1016/j.bspc.2019.101569>. ISSN 1746-8094.
- [35] A. Shankar, H.K. Khaing, S. Dandapat, S. Barma, Analysis of epileptic seizures based on EEG using recurrence plot images and deep learning, *Biomed. Signal Process Control* 69 (2021) 102854, <https://doi.org/10.1016/j.bspc.2021.102854>. ISSN 1746-8094.
- [36] A. Bhattacharyya, R. Pachori, A. Upadhyay, U. Acharya, Tunable-Q Wavelet Transform Based Multiscale Entropy Measure for Automated Classification of Epileptic EEG Signals, *Applied Sciences* 7 (4) (2017), <https://doi.org/10.3390/app7040385>. 385.
- [37] A. Zahra, N. Kanwal, N. ur Rehman, S. Ehsan, K.D. McDonald-Maier, Seizure detection from EEG signals using multivariate empirical mode decomposition, *Comput. Biol. Med.* 88 (2017) 132–141, <https://doi.org/10.1016/j.combiomed.2017.07.010>. ISSN 0010 4825.
- [38] U.R. Acharya, S.L. Oh, Y. Hagiwara, J.H. Tan, H. Adeli, Deep convolutional neural network for the automated detection and diagnosis of seizure using EEG signals, *Comput. Biol. Med.* 100 (2018) 270–278, <https://doi.org/10.1016/j.combiomed.2017.09.017>.
- [39] A. Omar, T. Abd El-Hafeez, Optimizing epileptic seizure recognition performance with feature scaling and dropout layers, *Neural Comput & Applic* 36 (2024) 2835–2852, <https://doi.org/10.1007/s00521-023-09204-6>.
- [40] D. Thara, B. Prema Sudha, F. Xiong, Epileptic seizure detection and prediction using stacked bidirectional long short-term memory, *Pattern Recognit. Lett.* 128 (2019) 529–535, <https://doi.org/10.1016/j.patrec.2019.10.034>.
- [41] K. Akyol, Stacking ensemble based deep neural networks modeling for effective epileptic seizure detection, *Expert Syst. Appl.* 148 (2020) 113239, <https://doi.org/10.1016/j.eswa.2020.113239>. ISSN 0957-4174.
- [42] R. Abiyev, M. Arslan, J.B. Idoko, B. Sekeroglu, A. Ilhan, Identification of Epileptic EEG Signals Using Convolutional Neural Networks, *Appl. Sci.* 10 (12) (2020) 4089, <https://doi.org/10.3390/app10124089>.
- [43] S. Raghu, N. Sriraam, Y. Temel, S. Vasudeva Rao, P.L. Kubben, EEG based multi-class seizure type classification using convolutional neural network and transfer learning, *Neural Netw.* 124 (2020) 202–212, <https://doi.org/10.1016/j.neunet.2020.01.017>. ISSN 0893-6080.
- [44] M. Zubair, M.V. Belykh, Kumar Naik M. Umesh, Mohammad Fareeda Madeen Gouher, Shani Vishwakarma, Shaik Rafi Ahamed, Gouher, Shani Vishwakarma, Shaik Rafi Ahamed, Kongara Ramanjaneyulu, Detection of epileptic seizures from EEG signals by combining dimensionality reduction algorithms with machine learning models, *IEEE Sens. J.* 21 (15) (2021) 16861–16869, <https://doi.org/10.1109/JSEN.2021.3077578>.
- [45] M. Geng, W. Zhou, G. Liu, C. Li, Y. Zhang, Epileptic seizure detection based on Stockwell transform and bidirectional long short-term memory, *IEEE Trans. Neural Syst. Rehabil. Eng.* 28 (3) (March 2020) 573–580, <https://doi.org/10.1109/TNSRE.2020.2966290>.
- [46] R. Yuvaraj, J. Thomas, T. Kluge, J. Dauwels, A deep learning scheme for automatic seizure detection from long term scalp EEG, 2018 52nd Asilomar Conference on Signals, Systems, and Computers, 2018, pp. 368–372. Pacific Grove, CA, USA.
- [47] T.T. Chowdhury, A. Hossain, S.A. Fattah, C. Shahnaz, “Seizure and non-seizure EEG signals detection using 1-D convolutional neural network architecture of deep learning algorithm.” 2019 1st International Conference on Advances in Science, Engineering and Robotics Technology (ICASERT), 2019, pp. 1–4. Dhaka, Bangladesh.
- [48] M. Varli, H. Yılmaz, Multiple classification of EEG signals and epileptic seizure diagnosis with combined deep learning, *Journal of Computational Science* 67 (2023) 101943, <https://doi.org/10.1016/j.jocs.2023.101943>. ISSN 1877-7503.
- [49] S. Mallick V. Baths, Novel deep learning framework for detection of epileptic seizures using EEG signals, *Front. Comput. Neurosci.* 18 (2024 Mar 21) 1340251, <https://doi.org/10.3389/fncom.2024.1340251>. PMID: 38590939; PMCID: PMC11000706.
- [50] T. Kaur, T. Kumar Gandhi, Automated diagnosis of epileptic seizures using EEG image representations and deep learning, *Neuroscience Informatics* 3 (3) (2023) 100139, <https://doi.org/10.1016/j.neuri.2023.100139>. ISSN 2772-5286.
- [51] Z. Georgis-Yap, M.R. Popovic, S.S. Khan, Supervised and unsupervised deep learning approaches for EEG seizure prediction, *J Healthc Inform Res* 8 (2024) 286–312, <https://doi.org/10.1007/s41666-024-00160>.
- [52] Ö. Türk, M.S. Özerdem, Epilepsy detection by using scalogram based convolutional neural network from EEG signals, *Brain Sci.* 9 (2019) 115, <https://doi.org/10.3390/brainsci9050115>.
- [53] H. Meng, T. Yan, F. Yuan, H. Wei, Speech emotion recognition from 3D log-Mel spectrograms with deep learning network, *IEEE Access* 7 (2019) 125868–125881.
- [54] Daniel S. Park, William Chan, Yu Zhang, Chung-Cheng Chiu, Barret Zoph, Ekin D. Cubuk, Quoc V. Le, SpecAugment: a simple data augmentation method for automatic speech recognition, *INTERSPEECH* (2019), <https://doi.org/10.21437/Interspeech.2019-2680>.
- [55] H. Kui, J. Pan, R. Zong, H. Yang, W. Wang, Heart sound classification based on log Mel-frequency spectral coefficients features and convolutional neural networks, *Biomed. Signal Process Control* 69 (2021) 102893, <https://doi.org/10.1016/j.bspc.2021.102893>.
- [56] A. Mukhamediya, S. Fazli, A. Zollanvari, On the effect of log-Mel spectrogram parameter tuning for deep learning-based speech emotion recognition, *IEEE Access* 11 (2023) 61950–61957.
- [57] H. Meng, T. Yan, F. Yuan, H. Wei, Speech emotion recognition from 3D log-mel spectrograms with deep learning network, *IEEE Access* 7 (2019) 125868–125881.
- [58] A. Meghanani, A. C.S. A.G. Ramakrishnan, An exploration of log-Mel spectrogram and MFCC features for Alzheimer’s Dementia recognition from spontaneous speech. 2021 IEEE Spoken Language Technology Workshop (SLT), 2021, pp. 670–677, <https://doi.org/10.1109/SLT48900.2021.9383491>. Shenzhen, China.
- [59] S. Wong, A. Simmons, J. Rivera-Villicana, S. Barnett, S. Sivathamboo, P. Perucca, et al., EEG datasets for seizure detection and prediction—a review, *Epilepsia Open* 8 (2023) 252–267, <https://doi.org/10.1002/epi4.12704>.
- [60] R.G. Andrzejak, K. Lehnertz, F. Mormann, C. Rieke, P. David, C.E. Elger, Indications of nonlinear deterministic and finite-dimensional structures in time series of brain electrical activity: dependence on recording region and brain state, *Phys. Rev. E: Stat. Phys., Plasmas, Fluids, Relat. Interdiscip. Top.* 64 (6) (2001) 061907, <https://doi.org/10.1103/PHYSREVE.64.061907>.
- [61] T.K. Gandhi, P. Chakraborty, G.G. Roy, B.K. Panigrahi, Discrete harmony search based expert model for epileptic seizure detection in electroencephalography, *Expert Syst. Appl.* 39 (2012) 4055–4062, <https://doi.org/10.1016/j.eswa.2011.09.093>.
- [62] P. Swami, M. Bhatia, M. Tripathi, P.S. Chandra, B.K. Panigrahi, T.K. Gandhi, Selection of optimum frequency bands for detection of epileptiform patterns, *Healthc. Technol. Lett.* 6 (2019) 126–131, <https://doi.org/10.1049/hlt.2018.5051>.
- [63] P. Swami, T.K. Gandhi, B.K. Panigrahi, M. Bhatia, J. Santhosh, S. Anand, A comparative account of modelling seizure detection system using wavelet techniques, *Int. J. Syst. Sci. Oper. Logist.* 4 (2017) 41–52, <https://doi.org/10.1080/23302674.2015.1116637>.
- [64] P. Swami, T.K. Gandhi, B.K. Panigrahi, M. Tripathi, S. Anand, A novel robust diagnostic model to detect seizures in electroencephalography, *Expert Syst. Appl.* 56 (2016) 116–130, <https://doi.org/10.1016/j.eswa.2016.02.040>.
- [65] Vinícius R. Carvalho, Márcio F.D. Moraes, Antônio P. Braga, Eduardo M.A.M. Mendes, Evaluating five different adaptive decomposition methods for EEG signal seizure detection and classification, *Biomed. Signal Process Control* 62 (2020) 102073, <https://doi.org/10.1016/j.bspc.2020.102073>. ISSN 1746-8094.
- [66] A. Mukhamediya, S. Fazli, A. Zollanvari, On the effect of log-Mel spectrogram parameter tuning for deep learning-based speech emotion recognition, *IEEE Access* 11 (2023) 61950–61957, <https://doi.org/10.1109/ACCESS.2023.3287093>.
- [67] P. Swami, B.K. Panigrahi, S. Nara, M. Bhatia, T.K. Gandhi, EEG epilepsy datasets. <https://doi.org/10.13140/RG.2.2.14280.32006>, 2016.